Legible Consensus: Capability Gaps in Flexible Quorums

Paper #98

Abstract

A correct quorum answer can still leave a service unable to do what its operators assume the answer covers: elect a new leader, or commit new writes. Classic majority Paxos hid this possibility: one threshold governed both acquiring proposal authority (Paxos Phase 1; leader election in Multi-Paxos) and committing values (Phase 2), so one reachable-replica count answered both questions. Flexible quorum systems keep the cross-phase intersection required for safety while letting the two thresholds diverge, and production systems adopt them precisely for that freedom. The consequence is documented operationally: the same reachable set can support acquisition but not commit, or commit but not acquisition, while the protocol remains safe.

This paper gives that split an exact boundary, a measured interior, and an executable recognizer. The boundary is containment: acquisition without commit is impossible if and only if every Phase 1 quorum contains a Phase 2 quorum (and symmetrically the other way), a design-time test for any pair of quorum families. Inside the boundary, a pre-registered contention experiment reversed our predicted severity ordering: the acquisition-only state we expected to be disruptive matched a healthy competitor on every recorded metric, while a commit-capable but acquisition-incapable incumbent with a matched round budget starved a healthy proposer in every seed. For recognition, a crumbling-wall construction shows that quorum shape can exclude chosen mixed states, and a readout linear in the number of tiers names which capabilities remain and which obligations failed, from configuration and observed connectivity alone; runtime authority and service policy stay outside what any structural test can supply.

Mars and LEO make prolonged and time-varying reachability impossible to ignore; changing membership itself remains outside this fixed-epoch result.

1 Introduction

At 22:52 UTC on October 21, 2018, routine work to replace failing 100G optical equipment interrupted connectivity between two parts of GitHub's US East Coast infrastructure. Connectivity returned 43 seconds later. During that interval, a surviving quorum of Orchestrator's Raft group initiated database failover, exactly as configured. The old primary site retained writes that had not reached the new one; after connectivity returned, the new primaries accepted further writes; and the application tier could not tolerate the latency of the resulting cross-country topology. Recovery and degraded service lasted 24 hours and 11 minutes [21]. Consensus answered the question it was asked: a quorum of reachable replicas agreed to fail over. The questions the operators needed answered were different: what can the service still do, and what will this action cost it? This paper is about the gap between the question consensus answers and the questions operators need answered.

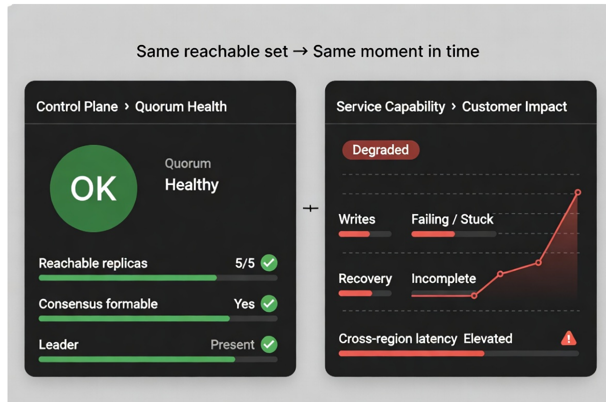

Figure 1: *The same reachable set can produce a healthy control-plane answer while the service remains unable to deliver the expected capabilities. Conventional majority Paxos made the two questions coincide; flexible quorums dissolve that coincidence while remaining safe.*

The problem we study is that a single health or quorum answer can conceal two different consensus capabilities. Given the replicas currently reachable, can a proposer acquire proposal authority (complete Phase 1; leader election in Multi-Paxos)? Can an authorized proposer commit? Flexible quorum systems can answer these questions differently, yet conventional health summaries do not identify which capability remains. We ask which mixed states a quorum design permits, how an operator can recognize the current state from the configured quorums and observed connectivity, and what that state does (and does not) say about recovery. The GitHub incident did not necessarily involve this specific mechanism; it illustrates the broader danger of treating a correct control-plane answer as evidence of every capability the service requires (Figure 1).

Meta's LogDevice supplies the narrower technical instance. LogDevice uses flexible Multi-Paxos: an incumbent leader can retain the ability to write after failures remove the larger quorum needed to elect and recover a replacement. Meta reports that the resulting stuck recoveries recurred at fleet scale, violated read-availability objectives, and required manual intervention until LogDevice's design was augmented [16]. Meta understood and engineered around this tradeoff. GitHub shows a correct control answer failing to establish service-level capability; LogDevice shows the acquisition/commit capability split that we analyze.

These are not exotic configurations. Phase asymmetry is exactly what buys local commit latency in WAN and edge deployments, so the same pressure that spreads flexible quorums also dismantles, quietly and deliberately, the coincidence that made a replica count a health answer.

Why had that split been so easy to overlook? Familiar majority-based Paxos configurations commonly use the same threshold quorum family for both phases. Phase symmetry makes two questions—can a proposer acquire proposal authority, and can it exercise that authority to complete a commit?—receive the same answer for every reachable set. A node-health dashboard answers neither. The threshold, however, makes their shared answer readable from one replica count at the network vantage under consideration. We call this equality of formability predicates *phase-capability coincidence*. The smaller question, "how many replicas are reachable?", appeared to answer both larger ones because the answers happened to coincide. Flexible Paxos [10] preserves the cross-phase intersection required for safety while allowing that coincidence to dissolve. The health or count answer did not become false; the larger inference from it stopped following.

Three things here are new. First, the boundary: for any finite pair of quorum families, containment tells us exactly which acquisition-without-commit and commit-without-acquisition states the pair admits, and a construction-independent auditor makes the test executable (Section 3). Second, the interior: a pre-registered, single-decree contention experiment measures behavior inside the two directions, and its reversal of our predicted ordering shows that direction and retry policy, not mere existence, determine a gap's operational cost (Section 4). Third, the recognizer: a crumbling-wall construction serves as an instrument, demonstrating that quorum choice can prevent selected gaps and providing a concrete $O(\text{tiers})$ readout of the capabilities and failed obligations that remain (Sections 5 and 6). We call a construction *legible with respect to a supplied connectivity summary* when it provides such a readout without enumerating candidate quorum subsets or attempting protocol execution. The phenomenon itself is not new: Flexible Paxos permits it by design, and Meta operated inside it deliberately. What was missing, and what this paper supplies, is the exact boundary, the measured interior, and the executable recognizer.

The mixed states carry no policy-independent verdict. LogDevice used retained commit capability under existing authority as an availability feature; under our single-decree contention policy, the same structural direction exhausted a competing proposer's progress budget. Detecting a gap is therefore insufficient: its direction, current authority, and service policy determine what an operator should do.

The resulting response has three layers. *Prevent*: choose quorum families whose containment excludes an unacceptable mixed state. *Detect*: audit the design and expose the phase capabilities permitted by supplied connectivity. *Act*: combine those structural answers with fresh runtime authority and an explicit service policy. The last layer is necessary and outside our model; the readout does not detect connectivity, establish current authority, select recovery policy, or guarantee that an operator consults it. Mars and LEO make long and time-varying reachability impossible to hide behind millisecond intuition. The harder case in which membership itself changes remains future work. Our results stay within fixed configuration epochs, and the 5/1/1/3 planetary system used below is an evaluated fixture rather than a proposed deployment.

We begin with the historical convenience that made the smaller question work.

2 Why One Count Once Worked

For one replica count to answer both operational questions, two facts must line up. Classic Paxos [12] commonly uses one majority-quorum family for Phase 1 (prepare/promise) and Phase 2 (accept/accepted). Because the families are the same, a reachable set that can form one phase can form the other. Writing $\mathrm{Form}(\mathscr{Q})$ for the connectivity sets that contain some quorum in family $\mathscr{Q}$, we call

$$\mathrm{Form}(\mathscr{Q}_1) = \mathrm{Form}(\mathscr{Q}_2)$$

phase-capability coincidence. Phase symmetry is sufficient for this equality, but not necessary: syntactically different families can induce the same formability predicate.

Majority supplies the second fact. A majority is a threshold family, so its formability can be decided by counting reachable replicas from the network vantage under consideration. The count is useful because the threshold makes the predicate readable. It answers both questions because phase symmetry has already made the predicates equal.

Flexible Paxos [10] separates the safety requirement from this convenient symmetry. Safety requires *cross-phase* intersection: every Phase 1 quorum must intersect every Phase 2 quorum. Quorums within one phase need not intersect one another. For uniform thresholds, $q_1 + q_2 > |N|$ is sufficient. A designer may therefore enlarge the less frequent Phase 1 family so that Phase 2, the commit hot path, remains small or local. Once the two families differ, the two operational questions can receive different answers. Intersection is the requirement; majority is a proof method, pigeonhole on cardinality, which makes the guarantee checkable by one comparison against one count. Uniform thresholds keep the count informative at two comparisons instead of one; shaped families retire it, because formability stops depending on how many replicas respond and starts depending on which.

For a Moon proposer under the strict wall used below (the construction of Section 6), two reachable sets of five nodes give opposite answers; R_1 asks whether the reachable set supports acquisition, R_2 whether it supports commit:

Reachable set	Count	R_1	R_2
All five Earth nodes	5	no	yes
Three Earth nodes, LEO, and Moon	5	yes	no

Moon's Phase 1 family needs one Earth node, LEO, and Moon. Strict Phase 2 needs all five Earth nodes. The same count therefore describes either mixed state. Five is no longer informative by itself because it omits the phase structure that distinguished the cases. Once phase-capability coincidence is lost, the smaller question no longer settles the two larger ones.

The uniform case shows the smallest cost of retaining the coincidence. Subject to $q_1 + q_2 \geq n+1$, a cost-minimal split can be symmetric when n is odd. When n is even, symmetry adds one additional participant across the two phase thresholds. This is a post-hoc structural observation, not an explanation for deployment practice or for the odd-cluster convention. Once the families differ, the remaining task is to determine which mixed states must appear and which quorum design can exclude.

3 Capability Gaps

The operator's two questions have a common form, and nothing in this section depends on the construction that follows: the definitions and results below hold for any pair of quorum families over a finite universe. For either phase, the plain-language question is whether the nodes currently reachable contain any configured quorum for that phase. The notation below records exactly that answer.

For a finite fixed universe N_e and a nonempty quorum family $\mathscr{Q}$, define its formability predicate by

$$\mathrm{Form}(\mathscr{Q}) = \{C \subseteq N_e \mid \exists Q \in \mathscr{Q} : Q \subseteq C\}.$$

Given phase families $\mathscr{Q}_1$ and $\mathscr{Q}_2$, write $R_1(C)$ when $C \in \mathrm{Form}(\mathscr{Q}_1)$ and $R_2(C)$ when $C \in \mathrm{Form}(\mathscr{Q}_2)$. We say a proposer can *acquire* when R_1 holds and is structurally commit-capable when R_2 holds. Actual commitment also requires valid runtime authority and ordinary protocol execution. Under Multi-Paxos, acquisition is leader election and amortizes over many commands; under single-decree Paxos it is per-decree ballot acquisition and every decree pays for it. Our measurements are single-decree throughout (Section 4), so "acquisition" carries the general sense everywhere except the design-time leadership reading in Section B.

The pair (R_1, R_2) has four joint states. Two are familiar: $(1,1)$ makes both phases formable, and $(0,0)$ makes neither formable. We call the mixed states *capability gaps*: $(1,0)$ permits acquisition but not commit formation, while $(0,1)$ permits commit formation but not acquisition. A single family used for both phases makes both gaps empty. Phase-capability coincidence is the semantic equality $\mathrm{Form}(\mathscr{Q}_1) = \mathrm{Form}(\mathscr{Q}_2)$; it can also hold for syntactically different families.

The ordering of the two formability sets supplies the entire classification. If every connectivity set capable of acquisition is also capable of commit, acquisition without commit has no witness, although commit without acquisition may remain. If neither formability set contains the other, each has a connectivity witness absent from the other, and both mixed directions occur. The containment condition below makes this predicate order checkable directly from the quorum families.

Proposition 1 (gap emptiness). The $(1,0)$ gap is empty if and only if every $Q_1 \in \mathscr{Q}_1$ contains some $Q_2 \in \mathscr{Q}_2$. Dually, the $(0,1)$ gap is empty if and only if every $Q_2 \in \mathscr{Q}_2$ contains some $Q_1 \in \mathscr{Q}_1$.

Proof. Both predicates are monotone in C: adding a reachable node never revokes formability. If some Q_1 contains no Q_2, take $C = Q_1$; then $R_1(C)$ holds and $R_2(C)$ fails. Conversely, if every Q_1 contains a Q_2, every C admitting a Q_1 admits that contained Q_2. The dual argument is symmetric. □

Corollary 1 (exact correspondence). The reachable gap profile is a complete invariant of the ordering between the two formability predicates: equal predicates admit neither gap; R_1 strictly implying R_2 admits only $(0,1)$; R_2 strictly implying R_1 admits only $(1,0)$; and incomparable predicates admit both. This follows by applying Proposition 1 in both directions and distinguishing equality from the two strict implications.

The proof is short because formability is monotone. While both phases used the same family, the question had nothing to distinguish: both gaps were empty by inspection. Flexible quorums made the question useful by allowing the phase predicates to differ. Proposition 1 then says exactly how they differ.

Our construction-independent auditor implements this test over explicit finite families. It validates the universe and families, removes duplicate and nonminimal quorums, checks every cross-phase pair for safety, classifies the four predicate relations, and emits deterministic minimum-cardinality gap witnesses. An optional pinned set restricts the analyzed domain to connectivity states containing those nodes without changing the configured families or their safety result.

Writing $\min(\mathscr{Q})$ for the canonical minimal antichain, classification takes $O(|\min(\mathscr{Q}_1)||\min(\mathscr{Q}_2)||N_e|)$ time, excluding input parsing, deterministic sorting, and the separate quadratic-in-family-size minimization passes. An independent mode enumerates every connectivity set rather than reusing the containment classifier. It rejects a deliberately mutated report and agrees with the optimized classifier over all 129,032 combinations of nonempty family pair and pinned domain on a three-node universe. The proof above remains primary; the auditor makes the criterion executable and checks the registered wall instances against the existing connectivity enumeration with zero disagreements.

The two mixed states are structurally symmetric in the characterization, but need not behave symmetrically. Before reading the next section, which would you expect to disrupt a competing healthy proposer: acquisition without commit, or commit capability without fresh acquisition? We registered the first prediction before implementing the experiment.

4 What Happens Inside the Gaps

The experiment uses a deliberately bounded fixture of ten acceptors: five on Earth, one in LEO, one on the Moon, and three on Mars. Strict Phase 2 requires all five Earth acceptors (the Phase 2 threshold $k = 5$); the Moon incumbent's Phase 1 requires the Moon and LEO acceptors plus at least one Earth acceptor. The full construction follows in the next major section. Here the fixture exists only to place an incumbent in each mixed state and observe its interaction with a healthy proposer; it is not a deployment proposal.

Before the experiment code existed, we registered four predictions, their falsification criteria, and the consequence of a null result. The primary prediction was that an acquisition-without-commit $(1,0)$ incumbent would delay a healthy proposer more than an incumbent that fails Phase 1. Overlapping Wilson intervals would falsify that prediction; a higher completion rate for the $(1,0)$ arm with disjoint intervals would reverse it. The remaining predictions concerned whether disruption was transient, whether repeated attempts created the expected cost, and whether the timing of the Phase 1 failure mattered.

The resulting single-decree Paxos experiment has five arms: a mid-round flip from $(1,1)$ to $(1,0)$, no incumbent, an incumbent fixed in $(0,1)$, a healthy $(1,1)$ incumbent, and a mid-round flip to $(0,1)$. A healthy Earth proposer competes with a Moon incumbent at Phase 2 threshold $k = 5$. The incumbent's maximum round count is swept over $\{1,2,4,8\}$, with 50 seeds per cell; the healthy proposer has a fixed eight-round budget. The harness records completion only when the healthy proposer returns success backed by a decision certificate. "Prevented decision" means that it did not do so before the run ended after the healthy proposer's fixed eight-round budget. The phrase is not evidence of livelock. The table collapses the five arms onto three rows: the flip-to-$(1,0)$ and healthy arms are constant across all incumbent budgets; the $(0,1)$ row merges the fixed and late-flip placements at incumbent budget eight, where their NACK counts differ by one; and the no-incumbent control (median first commit 0.182 s, one round, zero NACKs) is omitted as the uncontended baseline.

Incumbent condition	$P(\text{ok})$	med. ttfc	rounds	NACKs
Acquire-only $(1,0)$, b. 1,2,4,8	1.000	3.420 s	2	7
$(1,1)$ healthy, b. 1,2,4,8	1.000	3.420 s	2	7
Commit-only $(0,1)$, b. 8	0.000	—	8	48–49

The prediction posed at the end of the previous section failed (Figure 2). It failed in reverse, just as the dated preregistration permits us to say without reconstructing the surprise. At every incumbent budget, the $(1,0)$ arm matched the healthy $(1,1)$ contender on every recorded metric: completion probability 1.000, median time to first commit 3.420 s, two healthy rounds, and seven NACKs. The deterministic fixture produced the same recorded healthy-proposer trace in both arms, which explains the exact equality rather than merely similar medians. The incumbent completed from one to eight Phase 1 quorums as its budget increased, while the healthy outcome stayed flat. This is an experimental equivalence on the recorded metrics under these conditions, not a universal claim about $(1,0)$.

The $(1,0)$ incumbent also supplied the value that was decided. Its partial Phase 2 reached four of five Earth acceptors. The healthy proposer's next Phase 1 found that accepted value, and ordinary Paxos value adoption carried it to commitment. The result therefore demonstrates the recovery mechanism described for Flexible Paxos: acquisition can preserve and finish work begun by a proposer that lost commit capability.

The $(0,1)$ arms produced the other direction. With incumbent budgets one, two, and four, the healthy proposer completed in all 50 seeds, while median time to first commit rose from 3.420 to 6.665 to 13.186 s in the early arm. At incumbent budget eight, the healthy proposer exhausted its eight rounds in all 50 seeds; early and late placement both produced zero completions, with 48–49 NACKs. That outcome is the second registered falsification: the fixed $(0,1)$ arm's disruption was predicted to remain transient per round, and at budget eight it was total. These are bounded failures under one topology, one severed link, and the modeled backoff policy. They establish neither intrinsic harm nor livelock.

The traces expose the mechanism. A proposer that fails Phase 1 receives NACKs and returns to its short backoff. A proposer that completes Phase 1 but cannot form Phase 2 waits for the phase timeout before trying again. In this harness, success in the first phase slows the rate at which the incumbent can raise ballots. The timeout supplies a structural floor because it must exceed the round trip; the backoff policy determines how fast the floor-free cycle runs.

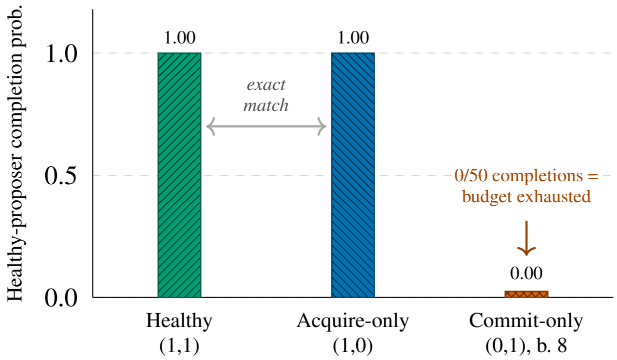

Figure 2: *Pre-registered single-decree contention experiment (50 seeds per cell). The prediction that an acquire-only* $(1,0)$ *incumbent would be more disruptive was reversed: it matched the healthy baseline on every recorded metric, while a commit-only* $(0,1)$ *incumbent with a round budget of 8, which is equal to the healthy proposer's own fixed budget, exhausted the healthy proposer's progress budget (0/50 completions, drawn as a minimal stub for visibility).*

Under Multi-Paxos, incumbent authority can hide missing acquisition capability: a still-authorized commit-without-acquisition $(0,1)$ leader can continue Phase 2 until authority must be reacquired. Single-decree Paxos cannot hide it because every decree pays for Phase 1, so this experiment exposes the state immediately. LogDevice used retained incumbent write capability in this structural direction as an availability feature [16]; in our bounded contention experiment, the same direction exhausted the healthy proposer's progress budget. Neither system is causal evidence for the other. Their opposite uses show that a structural state has no policy-independent valence. The operational response can therefore reverse with the state: restarting an authorized $(0,1)$ leader can discard the only currently usable authority, while acquisition through a $(1,0)$ state can recover an accepted value. A node-health summary distinguishes neither case.

Five deviations remain recorded with their reasons in the artifact: the flip occurs before Phase 2 fan-out; the intended $(0,0)$ control is realized as $(0,1)$; the incumbent is Moon-tier; the late-$(0,1)$ arm was added to test the timing prediction; and the registered treatment exists only at $k=5$. At $k=3$, the boundary result (Proposition 3) makes $(1,0)$ unreachable at every tier, so there is no corresponding treatment state to rerun. That result and the wall enumeration stand independently of this experiment. These runs measure behavior inside constructed mixed states; they do not validate either structural result. How can a designer or operator read which state the topology permits?

5 Look Through the Instrument

The repository's deterministic readout answers that question for a supplied configuration and connectivity summary. For the supplied Moon-initiated example, it prints:

```
state: (0,1)
initiating tier: Moon
R1 acquisition formable: False
R2 commit quorum formable: True
requires preexisting authority: True
runtime authority: unknown
service policy: not-inferred
missing Phase 1 LEO obligation: require 1, reachable 0
```

The two phase capabilities are visibly different. The Moon initiator cannot form Phase 1, but the reachable Earth nodes can form Phase 2. A scalar health summary would obscure that distinction; the two-bit state reports it directly.

The readout also names the failed obligation. It reports the missing LEO witness rather than requiring an operator to infer the problem from timeouts or repeated protocol attempts. This is structural interpretation of known connectivity, not failure detection or empirical evaluation.

Finally, the output refuses to invent what its inputs cannot establish. Commit formability in this state requires preexisting authority, but current authority is unknown and service policy is not inferred. The instrument says what the topology permits; it does not identify a current leader, establish fresh authority, choose a client contract or recovery policy, or ensure that anyone acts on the result.

The wall-specific pass takes named tiers, a Phase 2 threshold, an initiating tier, a reachable-node summary, and the provenance of configuration and connectivity. It returns (R_1, R_2), quorum witnesses, and typed failed obligations in $O(\text{tiers})$ time. The planetary input is a demonstration artifact, not empirical evidence. The edge input exercises the same interface as a structural mapping and is not an evaluated result; it supports no claim about terrestrial latency, availability, or prevalence. The next section presents the wall construction that generates these tier obligations and makes the compact pass possible.

6 Putting the Gaps on a Wall

Peleg and Wool [18] arrange nodes in rows of potentially different widths. A crumbling-wall quorum takes one complete row and one representative from every row below it. Two such quorums intersect: if their complete rows differ, the higher quorum carries a representative in the lower quorum's complete row; if the complete rows match, they share that row.

We split this geometry across the two Paxos phases. Each tier's Phase 1 family reads downward from that tier, while Phase 2 stays on the bottom row. Extra downward witnesses express participation policy; the Earth anchor supplies the Paxos intersection. Removing one of those witnesses changes that policy, not the anchor-based safety argument.

6.1 System Model

For each configuration epoch e, let N_e be the fixed logical acceptor universe and let $C_t \subseteq N_e$ be the acceptors effectively reachable at time t. A *decision window* is the interval over which we evaluate phase formability against one such connectivity state; N_e remains fixed within that window even when C_t changes between windows. Our results apply epoch by epoch.

Changing N_e is a reconfiguration and state-transfer decision that requires its own protocol. It is outside this paper's scope and is not modeled as ordinary reachability churn. For the evaluated epoch below, we write N for N_e.

Let the acceptor universe be

$$N = E \cup L \cup U \cup M,$$

where the tiers are pairwise disjoint and represent Earth, LEO, Moon, and Mars respectively. In the evaluated topology,

$$|E| = 5, \quad |L| = 1, \quad |U| = 1, \quad |M| = 3, \quad |N| = 10.$$

We call this the 5/1/1/3 topology. Tiers are ordered by latency: Earth is the fastest (bottom of the wall), Mars is the slowest (top). The network is asynchronous with configurable per-link delay and jitter. Acceptors are crash-stop and non-Byzantine. Links may be removed during scheduled blackout windows.

The simulator uses three logical consensus scopes:

- **Earth-local**: Flexible Paxos over the five Earth nodes ($q_1 = 4$, $q_2 = 2$).
- **Mars-local**: majority quorum over the three Mars nodes ($q = 2$).
- **Global**: the crumbling-wall construction defined below, over all ten nodes.

6.2 Per-Tier Phase 1 Families

The wall maps tiers to rows, ordered bottom (Earth, fast) to top (Mars, slow), as shown in Figure 3. We index tiers from the bottom: $T_0 = E$ (Earth), $T_1 = L$ (LEO), $T_2 = U$ (Moon), $T_3 = M$ (Mars). A proposer at tier i forms its Phase 1 quorum by reading *downward* from row i: it needs at least one respondent from its own tier and every tier below.

Initiating tier	Phase 1 scope	Tiers	Min size
Earth (tier 0)	Earth only	1	1
LEO (tier 1)	LEO + Earth	2	2
Moon (tier 2)	Moon + LEO + Earth	3	3
Mars (tier 3)	all four tiers	4	4

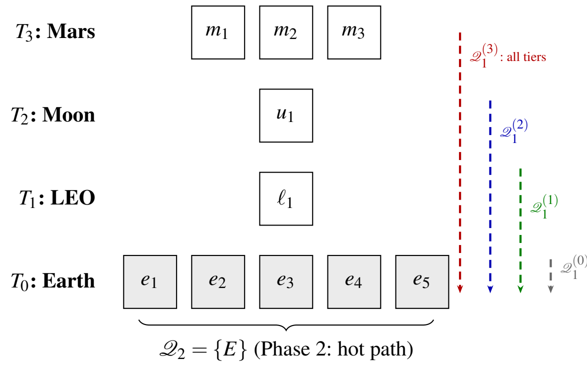

Figure 3: *The crumbling-wall construction over the 5/1/1/3 topology. Each tier's Phase 1 quorum reads downward through the wall (dashed arrows). Phase 2 uses only the Earth row (shaded). Row widths vary; wider rows offer more quorum choices.*

Let $\mathscr{Q}_1^{(i)}$ denote the Phase 1 family for a proposer at tier i:

$$\mathscr{Q}_1^{(i)} = \left\{ Q \subseteq N \,\middle|\, \forall j \leq i : \; Q \cap T_j \neq \emptyset \right\}.$$

A proposer at tier i needs at least one node from every tier at or below it ($j \leq i$). For example, $\mathscr{Q}_1^{(2)}$ (Moon) requires intersection with T_0 (Earth), T_1 (LEO), and T_2 (Moon), which is every tier from Moon down to Earth. For relaxed Phase 2 (Section 8.1), Phase 1 additionally requires enough Earth nodes to satisfy a pigeonhole intersection constraint.

Only the Earth floor is required for Paxos safety. Each additional downward-path witness represents a deployment policy that requires a tier to participate in acquisition, such as local representation, administrative control, sovereignty, or fencing before authority crosses a boundary. If no such policy applies, the extra witness is not justified by consensus itself; in that setting the wall should be read as an analytical construction rather than a recommended deployment geometry.

6.3 Phase 2 Family

The Phase 2 family places the hot commit path entirely on the fast tier:

$$\mathscr{Q}_2 = \{E\}.$$

That is, Phase 2 requires all five Earth nodes (the strict construction). Section 8.1 relaxes this to k-of-$|E|$.

6.4 Safety: Quorum Intersection

Proposition 2. For every tier i and every $Q_1 \in \mathscr{Q}_1^{(i)}$, $Q_2 \in \mathscr{Q}_2$: $Q_1 \cap Q_2 \neq \emptyset$.

Proof. Every $\mathscr{Q}_1^{(i)}$ requires $Q_1 \cap T_0 \neq \emptyset$ (since $0 \leq i$ for all i, i.e., Earth is at or below every tier). $Q_2 = E = T_0$. Therefore $Q_1 \cap Q_2 \supseteq Q_1 \cap T_0 \neq \emptyset$. □

The intersection follows from the wall's structure: every path through the wall reaches the bottom row, and Phase 2 is the bottom row.

We verified this mechanically in TLA+. For readers who have not used it: TLA+ is a specification language whose model checker, TLC, enumerates *every* state reachable from the initial conditions of a finite model and checks the stated invariant in each one. A run is *complete* when that enumeration reaches a fixed point and no reachable state remains unexamined. Such a run proves the invariant for that finite model; it is not a statistical sample of behaviors.

The `QuorumIntersection` specification enumerates every (construction $\times$ initiating tier $\times$ Q_1 $\times$ Q_2) combination for the strict, k=4, and k=3 families over the full 10-node topology (27,921 states, complete). `ExhaustiveIntersection` re-checks the strict and k=4 families as parse-time assumptions. No intersection failure was found.

For the full Paxos protocol (not just quorum intersection), we verify safety over a reduced 6-node topology ($3E + 1L + 1U + 1M$) that preserves the quorum-intersection structure. The invariant the reduction preserves is the pigeonhole inequality, not tier membership alone: in the reduced model both quorum families guarantee at least two of the three Earth nodes ($2+2>3$), and in the full modeled family Q_1 carries at least two Earth nodes against a 4-of-5 Q_2 ($2+4>5$). Both instantiate $|Q_1 \cap E| \geq |E| - k + 1$ with the margin $|E| - k = 1$ held fixed. (Tier membership alone suffices only under strict $\mathscr{Q}_2 = \{E\}$, where any Earth node in Q_1 intersects it.) Liveness properties (crash tolerance, progress under partial failure) are topology-size-dependent and are evaluated only at full scale. Over the reduced topology, single-decree Paxos agreement was model-checked completely (67M states, no violation).

Verification scope. The protocol-level specification models a single all-tier Phase 1 family rather than the per-tier families of Section 6: it exercises the protocol machinery under wall-shaped quorums, but safety of the per-tier construction itself rests on Proposition 2 and the standard Flexible Paxos argument, with the intersection premise checked exhaustively above. The proof is primary; the model checking is corroboration. Specifications are in `tla/` (see Appendix C).

6.5 What "Global" Means

When an Earth-based proposer completes both phases using only Earth nodes, the decision is *global* in the Paxos sense: any future proposer at any tier, upon completing Phase 1, will discover this value in the Earth acceptors' state and cannot contradict it. Global consensus does not require that all tiers participated, only that no future quorum can produce a conflicting result.

Safety establishes only the floor. The next section identifies which capability gaps the structure prevents.

7 Where the Wall Works, and Where It Stops

We report two domains of connectivity states. The *unconstrained* reading ranges over every reachable set; the *self-reachable* reading restricts that domain to sets containing the initiating tier's colocated acceptor. At $k=3$, the $(1,0)$ gap is absent at every tier: acquisition without commit has no reachable witness. At the same threshold, both gaps are absent for Earth under both connectivity readings, and both gaps are absent for LEO under the self-reachable reading. The commit path remains on Earth. The construction therefore restores the old coincidence at named tiers without returning inter-tier latency to Phase 2. The result has a boundary: $(0,1)$ remains reachable for Moon and Mars because their Phase 1 families retain downward participation obligations; this is commit without acquisition.

7.1 The Exact Boundary

Section 3 defines the two formability predicates for an arbitrary pair of quorum families. The wall instantiates them per tier: write $R_1(i,C)$ when $C \in \mathrm{Form}(\mathscr{Q}_1^{(i)})$ and $R_2(i,C)$ when $C \in \mathrm{Form}(\mathscr{Q}_2)$, so every tier carries its own acquisition predicate against one shared commit predicate. Which gaps are reachable is therefore a per-tier question, and the answer is not uniform across tiers.

Proposition 3 (boundary). For the construction of Section 6 under relaxed $\mathscr{Q}_2 = k$-of-$|E|$ (Section 8.1), $(1,0)$ is unreachable at *every* tier if and only if $|E| - k + 1 \geq k$, that is $k \leq \lceil |E|/2 \rceil$. Sufficiency is unconditional: it does not depend on the number of tiers, their sizes, or the initiating tier. The converse additionally requires non-degeneracy: no empty wall row.

The arithmetic is the same hitting-set bound that carries safety. A Q_1 holds at least $|E| - k + 1$ Earth nodes and a Q_2 holds k; when $|E| - k + 1 \geq k$, every Q_1 already contains a full Q_2 and Proposition 1 applies.

The gradient. Proposition 3 governs $(1,0)$ through the Earth floor. The $(0,1)$ direction follows a different mechanism: the downward chain. A proposer at T_2 (Moon) needs one witness from T_1 (LEO), and this topology has exactly one LEO node. Sever it while all five Earth nodes remain reachable and $\mathscr{Q}_2$ is formable while $\mathscr{Q}_1^{(2)}$ is not. Exhaustive enumeration over all 2^{10} connectivity states, for every tier and every k, gives the following compact table, listing the thresholds at which each gap remains reachable, under the self-reachable reading:

	T_0 (E)	T_1 (L)	T_2 (U)	T_3 (M)
Acquire-only $(1,0)$	$k \geq 4$	$k \geq 4$	$k \geq 4$	$k \geq 4$
Commit-only $(0,1)$	$k \leq 2$	$k \leq 2$	all k	all k

For $k \leq 3$, $(1,0)$ is absent at every tier. For $k \geq 3$, $(0,1)$ is absent for Earth under both readings and for LEO under the self-reachable reading. The LEO result depends on its initiator being able to reach its colocated acceptor. Under the unconstrained reading, LEO retains $(0,1)$ for every k: its single node can be severed while Earth remains whole. We state that sensitivity because it determines whether the coincidence extends to one tier or two.

Moon and Mars retain $(0,1)$ under both readings for every k. Each has an intervening witness row between itself and the Earth anchor, and changing the Earth threshold cannot remove that obligation. At the other endpoint, $k \geq 4$ violates Proposition 3's bound and opens $(1,0)$ everywhere. The useful design point is $k = 3$: it closes $(1,0)$ everywhere, closes both gaps at Earth under both readings and at self-reachable LEO, and keeps Phase 2 on Earth. No tested threshold closes both gaps at every tier.

The witnesses implement participation policy; legibility makes the resulting obligation readable. In the three-layer response, containment *prevents* selected gaps, the readout *detects* which residual structural state supplied connectivity permits, and authority plus service policy determine how to *act*. That last decision remains outside the interface. Enumeration and auditor artifacts are in `results/capability/` (see Appendix C). The remaining cost appears when failures reach the Earth anchor.

8 Prevention Has a Cost

Containment can prevent a capability gap, but it moves participation cost between phases. The exact boundary above identifies which mixed states disappear; the crash-relaxation experiment below shows the price of changing that boundary. Secondary calibration: the quorum-count gradient, competitive-majority baseline, full per-tier latency results, and sparse-topology sweep, which appears in Appendix B.

The calibration also separates wall position from convergence cost: in the modeled full-coverage topology, LEO completed faster than the cross-continental Earth path because its configured links were shorter. This is a fixture-specific result, not a claim about a realistic LEO deployment.

Configuration and connectivity remain separate inputs. A wall obligation may be satisfiable in the abstract while sparse reachability prevents an initiator from reaching the required Phase 2 nodes. The readout must therefore interpret both rather than infer one from the other.

8.1 Crash Tolerance and Coordinated Relaxation

The strict Phase 2 choice $\mathscr{Q}_2 = \{E\}$ introduces a fragility: a single crashed Earth acceptor blocks all global Phase 2 progress. This section relaxes that choice.

Relaxed construction. Let the relaxed Phase 2 family require k of $|E|$ Earth nodes. We evaluate $k = 4$ (tolerates one crash) and $k = 3$ (tolerates two). Phase 1 requires correspondingly more Earth nodes to maintain intersection: a k-of-$|E|$ Phase 2 quorum omits at most $|E| - k$ Earth nodes, so any set of $|E| - k + 1$ Earth nodes intersects every Phase 2 quorum (pigeonhole). Phase 1 must therefore contain at least $|E| - k + 1$ Earth nodes. For $k = 4$ out of 5 Earth nodes, this means at least 2 Earth nodes in every Q_1; for $k = 3$, at least 3. Both relaxed constructions are verified exhaustively in TLA+ alongside the strict one (Section 6).

The relaxed Phase 1 family for tier i is:

$$\mathscr{Q}_{1,k}^{(i)} = \{Q \subseteq N \mid \forall j \leq i : Q \cap T_j \neq \emptyset, |Q \cap E| \geq |E| - k + 1\}.$$

Table 1 reports the crash-tolerance sweep (Earth-initiated, repeater-assisted, 186 s Mars delay, 900 s blackout, 50 seeds).

With strict Phase 2, one Earth crash prevents every subsequent global commit (row 2). At $k = 3$ with two crashes, the global construction maintains 100% during-blackout success, while the standard Earth-local family cannot form its four-node Phase 1 quorum. Its 60.5% aggregate rate counts attempts completed before the crashes landed (rows 5–6). The weakest link migrates. For this Earth initiator, both relaxed global phases need the three surviving Earth nodes; the local family still requires four. Relaxing that local family to majority, $q_1 = q_2 = 3$, restores 100% local success while global success remains 100% (row 6). The tradeoff curve shows the corresponding Phase 1 minima for Earth and Mars initiators:

Regime	P1 min E/M	P2	Tol.	E-local $\mathscr{Q}_2$
Strict ($k = 5$)	1 / 4	5-of-5	0	2-node E
Relaxed ($k = 4$)	2 / 5	4-of-5	1	2-node E
Relaxed ($k = 3$) + maj	3 / 6	3-of-5	2	3-node E

An Earth-initiated Q_1 needs only the $|E| - k + 1$ Earth nodes; a Mars-initiated Q_1 adds one witness from each of the three upper tiers. Relaxation buys crash tolerance by growing the cold path, never the hot one: Phase 2 shrinks as Phase 1's Earth requirement grows.

The construction separates an inter-tier witness from intra-tier replication. Any one of the three Mars nodes satisfies the wall's Mars-row obligation; a Mars-local majority would require two. Earth is different because the Phase 2 anchor must act as one logical entity, whether by a local replicated state machine or another internal protocol. The reduced TLA+

Table 1: *Coordinated relaxation under Earth crashes (5/1/1/3; Earth initiator; repeater-assisted hard blackout; Mars-delay/blackout/timeout 186/900/500 s; 50 seeds; mean ± 95% CI). Earth-local Flexible Paxos: "std" is q_1=4,q_2=2; "maj" is q_1=q_2=3. Recovery is quantized by the 120 s reconciliation cadence.*

Global $\mathscr{Q}_2$	Local	Crashes	Earth-local	Global during	Global post	Recovery (s)
strict	std	0	100.0%	100.0%	100.0%	93.11±0.01
strict	std	1	100.0%	0.0%	0.0%	n/a
4-of-5	std	0	100.0%	100.0%	100.0%	94.27±0.01
4-of-5	std	1	100.0%	100.0%	100.0%	94.26±0.01
3-of-5	std	2	60.5%	100.0%	100.0%	94.64±0.01
3-of-5	maj	2	100.0%	100.0%	100.0%	94.63±0.01

model preserves this separation: shrinking Earth from five nodes to three and rescaling its threshold keeps the same intersection margin. Redundancy within a non-anchor tier therefore need not add another consensus step to the wall.

9 Related Work

Paxos established the standard safety argument for replicated state machines under asynchronous messaging and crash failures [12]. Flexible Paxos showed that safety requires only Phase 1/Phase 2 cross-intersection, not universal majority [10]. Our characterization builds on that result but does not claim containment itself as a new mathematical tool. The contribution is the exact correspondence between phase-formability ordering and its complete gap profile, the operational interpretation of those gaps, and their application to phase-asymmetric Paxos.

LogDevice provides an operational precedent for the commit-capable/acquisition-incapable direction. Meta deliberately retained incumbent write availability after losing the larger election and recovery quorum, then augmented the design after stuck recovery became a recurring source of read-SLA violations [16]. We do not claim discovery of that state. Our contribution is to characterize both directions for arbitrary finite quorum-family pairs, measure their behavior under one registered model, and provide executable recognition of the resulting predicate relation and witnesses.

The $(1,0)$ state is also present in Flexible Paxos: its $|Q_1|=1$, $|Q_2|=N$ construction permits acquisition before commit is formable, and its row-failure discussion uses Phase 1 to recover past decisions before reconfiguration. That work identifies the state and a constructive use for it. We ask the design-time question of whether a given construction admits the state at all, examine the dual state, and supply a readout for known connectivity. The readout matters because a recovery prescription must first know which predicate holds.

Containment has important prior uses. Refined Quorum Systems [8] nests quorum classes $QC_1 \subseteq QC_2 \subseteq RQS$ to connect resilience with optimistic complexity, subject to adversary-aware intersection properties. That class inclusion is a design and correctness condition, not an ordering of two phase-formability predicates over connectivity states. Li, Chan, and Lesani's quorum subsumption [13] is a member-indexed containment condition: for every process p in a quorum q, some quorum in p's personal family is contained in q. It supports progress in a heterogeneous Byzantine trust model. We instead compare two phase predicates over one fixed crash-stop acceptor universe. The quantifiers and the questions differ even though both results use containment.

Grid and crumbling-wall quorum constructions [3, 18] demonstrate that quorum geometry can be engineered around combinatorial structure. Howard et al. note explicitly that Flexible Paxos can use such constructions; we take up that suggestion. Peleg and Wool analyzed crumbling-wall availability under random, independent node failures. Our wall case study maps rows to physical latency tiers and examines per-tier liveness under scheduled, topology-correlated disconnection. Li, Chan, and Lesani's Satrapy system [14] carries heterogeneous quorum theory into practical consensus. Hierarchical quorum structures have a longer history in database replication [5] and coordination services; our wall differs in that the hierarchy reflects physical latency tiers rather than organizational or logical grouping.

Geo-distributed consensus provides the closest practical analog. WPaxos [1] uses Flexible Paxos with grid quorum layouts and per-object multi-leader coordination to keep commits local; it is the nearest point of comparison. WPaxos's grid quorums are symmetric: Q_1 spans $Z - f_z$ zones and Q_2 spans $f_z + 1$ zones, with the same structure regardless of which node initiates. Our per-tier $\mathscr{Q}_1^{(i)}$ families are asymmetric as the initiating tier determines the quorum scope, which produces a leadership cost gradient that symmetric grids structurally cannot express (Section B). EPaxos [17] removes the leader bottleneck for non-conflicting commands. Mencius [15] distributes leader responsibility across WAN sites. MDCC [11] achieves single-round cross-datacenter commits. All four optimize for tens-to-hundreds of milliseconds WAN latency and assume all replicas are usually reachable. Our setting differs in two respects: the asymmetry spans three orders of magnitude (sub-second to 1342 s), and the target phenomenon is scheduled total disconnection rather than heterogeneous latency.

Delay-tolerant networking [2, 7] motivates the operating regime but addresses routing and custody transfer, not consensus. CRDTs have been studied in DTN-like settings [9]. SwarmDAG [20] uses extended virtual synchrony for partition-tolerant robot swarms; Tran et al. [19] measure consensus latency under partitioning. Protocol Labs' IPC [6] uses hierarchical subnets for blockchain scaling. NASA's DSN study [4] evaluates database replication for ground stations. Our work occupies a gap: Paxos safety under topology-shaped quorum geometry and scheduled disconnection, with per-tier liveness characterization.

10 What Remains Unsolved

Mars makes one temporal regime impossible to ignore: logical membership can remain fixed while scheduled geometry removes reachability for days. The wall and auditor reason about this regime one configuration epoch and one supplied connectivity state at a time.

LEO suggests a faster version of the same problem. Logical membership may remain fixed while visibility changes on orbital timescales, making the sequence of connectivity states as important as any one snapshot. Our formability result classifies each snapshot; it does not establish that authority or accepted state remains usable across the transitions between them.

Mobile edge crosses the harder boundary. There, reachability varies while the authority-bearing membership itself may join, leave, or move between administrative domains. Treating that churn as a succession of fixed epochs is safe only if a reconfiguration and state-transfer protocol establishes how one epoch hands evidence to the next.

The present result therefore stops at fixed-epoch formability. It neither proposes a mobile Paxos protocol nor proves continuous availability across changing configurations. Scoped authority, gap-aware behavior, and multi-anchor families are possible directions, not results.

The open question is temporal: what overlap between successive quorum families, or what transferable evidence of authority and accepted state, is sufficient to preserve safety while membership and reachability change together? Mars exposes the duration of lost connectivity; LEO exposes changing visibility; mobile edge asks whether the configuration can change without losing the evidence the next configuration needs.

A compact framing for that question: the authority structure of reconfigurable Multi-Paxos is a hierarchical clock, $\langle$configuration, leadership, proposal$\rangle$, each counter meaningful only inside the frame above it, kept totally ordered by the protocol's hardest machinery. A reconfiguration is a tick of the slowest counter, and the open problem is what evidence must cross that tick so that two frames never run concurrently, each believing itself current. The introduction's incident shows the cost of such a fork in a system with no merge function; characterizing the evidence that prevents it is where this work points next.

11 Threats to Validity and Limitations

All results are design-level. The simulator evaluates one 5/1/1/3 crash-stop topology. Blackout is deterministic link removal; orbital motion, antenna schedules, contention, coding, variable link quality, Byzantine behavior, reconfiguration, and storage failure are absent. The fixed 120 s reconciliation cadence affects recovery point estimates, as stated beside those results. The crash sweep uses an Earth initiator and does not measure how wall position changes crash tolerance.

GitHub motivates the failure-backwards method: a correct control action can leave the service without capabilities its recovery requires. The containment theorem is not a root-cause analysis of that incident. LogDevice demonstrates the phase-capability split in a production-derived flexible Multi-Paxos design, but its recovery mechanism, read semantics, and incumbent authority differ from our Eidolon simulator's single-decree contention experiment. Neither production account validates the experiment, and the experiment does not explain either production system.

The behavioral experiment uses single-decree Paxos, one severed link, and one backoff policy. Multi-Paxos can defer exposure to a missing Phase 1 capability while existing authority remains valid. The experiment therefore measures one severe and deliberately bounded setting. Its results do not assign intrinsic valence to either mixed state.

The readout assumes supplied connectivity and returns structural formability, not runtime state: it detects no failure, establishes no fresh runtime authority, and supplies neither a chosen client contract nor recovery policy. The service policy and operator action remain outside the interface. The wall's linear readout also depends on one anchor tier; multi-anchor families may change both resilience and legibility.

We make no deployment prevalence claim and have no terrestrial evaluation. The edge input is a structural demonstration. A "tier" also bundles latency, failure domain, and administrative domain in the planetary case; those properties may separate elsewhere. Generalizing the structure is a mathematical possibility. Generalizing the measurements requires new experiments.

12 Conclusion

The smaller question worked because familiar phase-symmetric threshold quorums supplied phase-capability coincidence: acquisition and commit formability were equal, and their common answer was readable from one count.

Flexible quorums preserve the cross-phase intersection required for safety while dissolving that inference. Majority was the cheap-to-see quorum family; flexible quorums keep the requirement and repeal the cheap seeing. The count need not be wrong; it no longer answers both questions.

Containment gives designers a prevention boundary: it identifies exactly which mixed capability states a finite quorum-family pair excludes. The auditor and wall readout support detection by naming the capabilities and failed obligations that supplied connectivity permits. Action remains a third problem. It requires fresh runtime authority and an explicit service policy, neither of which structural formability can invent.

The changed practice is therefore to ask more than whether the system or its replicas are healthy. Ask which actions the system can presently complete, what authority each action requires, and what evidence supports each answer. That diagnostic applies within a fixed configuration epoch. The next problem begins when the system must carry the evidence itself across changing visibility and changing membership.

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Table 2: *Simulator parameters.*

Parameter	Value	Notes
Earth cross-cont. delay	30–120 ms	per link pair
LEO–ground delay	20–45 ms	per ground station
Moon one-way delay	1.28 s	fixed
Mars one-way delay	186–1342 s	swept
Mars-local delay	5 ms	fixed
Processing delay	1 ms	per acceptor
Jitter	$\pm 10\%$	of link delay, uniform
Per-phase timeout	500 s / 1 s	global / tier-local; global budget raised to $1.25\times$ the worst-case request–response path when that exceeds the base (above $\approx$200 s Mars delay)
Max Paxos rounds	1	per attempt
Blackout start	600 s base	pre-window scaled to fit one full global round plus margin (930 s at 186 s Mars delay)
Simulation end	4000 s base	horizon extends when pre/post windows scale
Reconciliation interval	120 s	between global attempts
Loss model	none	deterministic link removal for blackout

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A Simulator Parameters

B Supplemental Wall Calibration

Design-time gradient. The wall also makes the number of valid Phase 1 quorums readable by tier. If f_i nodes lie above

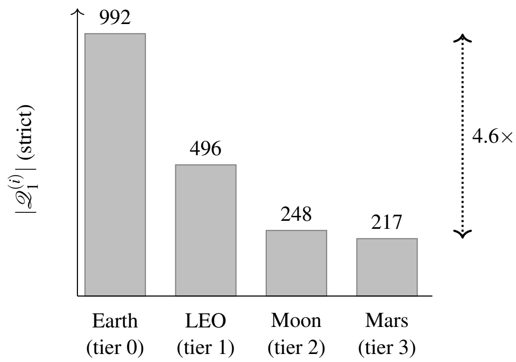

Figure 4: *Phase 1 quorum count by initiating tier under the strict wall. The values are exhaustive structural counts over the fixed 5/1/1/3 configuration; no connectivity distribution, timeout, or stochastic trial is assumed.*

tier i, then

$$N_i = 2^{f_i} \prod_{j \le i} (2^{|T_j|} - 1).$$

Each required row contributes a nonempty subset; nodes above the initiator are unconstrained. Moving up one tier changes exactly one factor from $2^{|T_{i+1}|}$ to $2^{|T_{i+1}|} - 1$, so $N_{i+1} < N_i$ for every wall. The 5/1/1/3 construction gives $992, 496, 248, 217$ from Earth through Mars (Figure 4). These are structural counts, confirmed by exhaustive enumeration; they are not availability estimates. An initiator-independent majority family assigns every tier the same 386 valid quorums and cannot express this gradient.

B.1 Evaluation Method

Eidolon is a discrete-event simulator built on SimPy. Acceptors process Prepare and Accept messages with configurable processing delay. Proposers fan out messages to all acceptors and collect responses with a per-phase timeout. The network model assigns entities to named locations with per-link latency, jitter, and optional loss. Blackout is modeled as deterministic link removal; the repeater model substitutes a degraded alternate path. Event ordering is by simulation timestamp; the simulator is deterministic given a seed.

Topology.

Five Earth ground stations (NA-West, Europe, Asia, SA-East, Africa) with cross-continental latencies of 30–120 ms. One LEO satellite with 20–45 ms links to ground stations. One Moon node at 1.28 s one-way. Three Mars nodes at configurable one-way delay (186–1342 s, representing closest approach to opposition). Mars-local links are 5 ms.

We evaluate two network variants:

- **Sparse**: LEO has links to 3 of 5 Earth stations (NA-West, Europe, Asia). Mars has links to 2 of 5 (NA-West, Europe). This models a single LEO satellite with limited ground coverage and a Mars relay with two tracking stations.

Table 3: *Experimental design.*

Experiment	Purpose	Key parameters
Parameter sweep	Confirm structural invariance of Earth-initiated global consensus across Mars delay and blackout duration	Mars delay $\in \{186, 750, 1342\}$ s; blackout $\in \{300, 900, 1800\}$ s; 50 seeds per point
Per-tier sweep	Measure which tiers retain global liveness; compare sparse vs. full-coverage topologies	Initiating tier $\in$ {Mars, Moon, LEO, Earth}; both topologies; 186 s Mars delay; 900 and 1800 s blackouts as stated; 50 seeds
Crash tolerance	Characterize the tradeoff between Phase 2 relaxation (k-of-$\|E\|$) and crash tolerance	$k \in \{3, 4, 5\}$; 0–2 Earth crashes; 186 s; 900 s blackout; 50 seeds

- **Full coverage**: LEO and Mars have links to all 5 Earth stations. This models a LEO constellation with global coverage and a deep-space network with worldwide antenna placement.

Experiments.

Three experiment families probe the construction. The separate registered experiment in Section 4 measures proposer behavior inside the two mixed states.

Fixed parameters across all experiments are listed in Table 2 (Appendix A). All seeds are 40–89 (50 runs per point). Results are reported as means with 95% confidence intervals.

B.2 Geometry and Competitive Majority Baseline

To calibrate the effect of quorum geometry, we compare three constructions over the identical topology, blackout model, and parameter sweep. The *flat* construction uses a single Phase 1 family requiring all four tiers:

$$\mathscr{Q}_1^{\text{flat}} = \{Q \subseteq N \mid \forall j \in \{0,1,2,3\} : Q \cap T_j \neq \emptyset\}.$$

Any blackout that removes a tier makes this quorum unformable. The *majority* baseline is a competitive shape-free Flexible Paxos configuration: $\mathscr{Q}_1^{\text{maj}} = \{Q \mid |Q| \geq 6\}$, any majority of the ten nodes, with no tier structure. The *crumbling wall* construction uses the per-tier Phase 1 families defined in Section 6, with an Earth-based global proposer. All three satisfy cross-intersection with $\mathscr{Q}_2 = \{E\}$: the tiered families require $Q_1 \cap T_0 \neq \emptyset$ directly, and any 6 of 10 nodes must include an Earth node because only five non-Earth nodes exist.

The flat construction yields 0% during-blackout success at every sweep point because it requires all tiers, including disconnected Mars. The crumbling wall yields 100% because the Earth-based proposer's Phase 1 reads only Earth. Zero variance across 50 seeds confirms that outcomes here are structural consequences of quorum geometry and link state, not scheduler artifacts. We therefore treat this as calibration, not discovery.

Table 4: *Effect of quorum geometry over the same 5/1/1/3 sparse topology. Earth-initiated proposer, hard blackout, timeouts as in Table 2, 50 seeds per point; rates are identical at all 9 parameter points (3 Mars delays × 3 blackout durations). Latency is a full two-phase single attempt.*

Construction	During	Post	Latency
Flat Phase 1 (all tiers required)	0.0%	varies	varies
Majority Phase 1 (any 6 of 10)	100.0%	100.0%	0.361 s
Crumbling wall (Earth reads Earth only)	100.0%	100.0%	0.183 s

Table 5: *Per-tier global consensus during Mars conjunction blackout (5/1/1/3 full-coverage topology, hard blackout, 186 s Mars delay, 1800 s blackout, 500 s phase timeout, 50 seeds). Rates are identical across seeds; latency and recovery are means, with 95% CIs reported in the text. Mars completes no during-blackout attempt, so its latency and recovery are post-blackout means; the inner tiers' values are during-blackout.*

Tier	Position	During	Latency	Rec. lag
Earth	bottom (0)	100.0%	0.183 s	9.4 s
LEO	tier 1	100.0%	0.131 s	8.1 s
Moon	tier 2	100.0%	5.131 s	3.0 s
Mars	top (3)	0.0%	753.2 s	1351.4 s

The majority baseline separates geometry from quorum slack. It also sustains 100% during-blackout success for the Earth-initiated proposer: the blackout removes three nodes and a 6-of-10 quorum tolerates four. Its two-phase decision latency is 0.361 s instead of 0.183 s. The sixth-fastest respondent lies outside Earth, so majority Phase 1 leaves the fast tier even when nothing has failed. Both constructions use the same all-Earth Phase 2; the difference appears in authority acquisition, recovery, and single-decree decisions. A LEO proposer in the sparse topology reaches only five acceptors during blackout, leaving majority Phase 1 unformable while the wall's LEO Phase 1 family remains formable. The majority family is insensitive to the initiating tier; Figure 4 shows the per-tier gradient that such a family cannot express.

B.3 Per-Tier Liveness under Full Coverage

Table 5 reports the central result under one observed condition. Each tier runs its own global proposer with full network coverage, 186 s one-way Mars latency, an 1800 s blackout, the hard-blackout model, and 50 seeds.

Earth, LEO, and Moon completed every during-blackout attempt in all 50 seeds. Mars completed none. The liveness failure crumbles from the disconnected top tier. The 95%

CIs are below ± 0.001 s for inner-tier latency and ± 0.02 s for inner-tier recovery; the Mars intervals are ± 0.82 s and ± 1.64 s, respectively.

The latency column maps directly to physics. Moon's 5.131 s is two Paxos phases at the 1.28 s one-way Moon–Earth light delay ($\approx 4 \times 1.28$ s). Earth's 183 ms reflects cross-continental round-trips within the Earth tier. LEO's 131 ms is *faster* than Earth, which is a counterintuitive result explained by the network topology: LEO-to-ground links (20–45 ms) are shorter than cross-continental Earth links (50–120 ms). Wall position determines quorum *obligations*; the speed of light determines their *cost*. These are correlated but not identical, and LEO is the proof.

Mars fails during blackout for the structural reason predicted by the wall: it cannot reach the inner-tier witnesses required for Phase 1. The 1800 s window contains complete failed attempts, so the measured during-blackout rate is 0% in all 50 seeds. Outside blackout, every post-blackout attempt completes, at a mean latency of 753.2 s. That is approximately two Paxos phases at the 186 s one-way Mars–Earth delay. Success requires a per-phase timeout budget sized to the initiating tier's physics: the worst-case request–response path from Mars is $2 \times 186 \approx 372$ s, within the 500 s per-phase budget of Table 2. Mars-initiated global consensus is possible outside blackout, but it pays the interplanetary path on both phases.

Recovery lag is 3.0 s for Moon, 8.1 s for LEO, 9.4 s for Earth, and 1351.4 s for Mars under this condition. These point values are cadence-alignment artifacts, not protocol guarantees. Recovery lag measures the gap from blackout end to the completion of the next successful attempt. Attempts are serialized: each launches one 120 s reconciliation interval after the previous attempt completes or times out, and an attempt already in flight when blackout ends must first exhaust its phase timeout, because its messages were dropped at send time. The lag is therefore bounded by one phase timeout plus one reconciliation interval plus the tier's round time (for Mars, $500 + 120 + 753 \approx 1373$ s; the measured 1351.4 s falls within it). Where the value lands inside that bound depends on where blackout end falls in the attempt sequence.[1]

B.4 Wall Obligations versus Sparse Reachability

Table 6 repeats the per-tier experiment under the sparse network topology, at the 900 s blackout condition (the full-coverage run of Table 5 used 1800 s).

LEO drops from 100% (full coverage) to 0% (sparse). The wall says LEO needs LEO + Earth; that obligation is satisfiable. But strict Phase 2 requires all five Earth nodes, and LEO has links to only three ground stations. The wall determines what a tier *needs*; the network determines what it *can reach*. Liveness requires both.

Table 6: *Per-tier global consensus under the 5/1/1/3 sparse topology (hard blackout, 186 s Mars delay, 900 s blackout, 500 s phase timeout, 50 seeds). All rates are identical across seeds; structural outcomes are reported without CIs.*

Initiating tier	Wall prediction	Sparse result
Earth	works	100.0%
LEO	works	0.0%
Moon	works	100.0%
Mars	blocked	0.0%

The readout therefore needs both supplied inputs. Configuration determines the obligations; connectivity determines whether the initiator can satisfy them. Full wall structure is insufficient when sparse reachability removes a required path.

Moon succeeds in both topologies because it has links to all five Earth ground stations. The sparse topology models a single LEO satellite with limited ground coverage; the two missing ground-station links are the difference between 0% and 100% for LEO-initiated consensus.

C Claim-to-Artifact Traceability

Table 7 maps the paper's main claims to executable artifacts. Paths are relative to the repository root of the (anonymized) artifact accompanying this submission. Results were generated with Python $\geq$3.14 and SimPy $\geq$4.1.1.

[1] Each tier's per-round time shifts its cadence phase relative to blackout end, which is why the values differ across tiers and move when the experiment window shifts. The effect is deterministic and independent of the consensus protocol; the bound, not the value, is the protocol property.

Table 7: *Claim-to-artifact traceability.*

Claim	Evidence	Artifact(s)
Flat construction: 0% during blackout at every hard-blackout sweep point.	Flat-mode sweep under the validated harness (`--global-quorum flat`).	`results/step9_flat/step9_sweep.csv`; `experiments/step9_sweep.py`.
Majority baseline: 100% during blackout, 361 ms two-phase decision latency at every hard-blackout sweep point.	Majority-mode sweep under the validated harness (`--global-quorum majority`).	`results/step9_maj/step9_sweep.csv`; `experiments/step9_sweep.py`.
Crumbling wall: 100% during blackout at every hard-blackout sweep point, 183 ms two-phase decision latency.	All rows show 100% during-blackout.	`results/step9/step9_sweep.csv`; `experiments/step9_sweep.py`.
Per-tier liveness: Earth, LEO, and Moon succeed; Mars fails under the common 1800 s condition.	Full coverage, hard blackout, 186 s Mars delay, 1800 s blackout, 50 seeds.	`results/tier_liveness/tier_sweep_full_ci.csv`; `experiments/tier_liveness_sweep.py`.
Sparse topology: LEO drops from 100% to 0%.	Sparse 5/1/1/3; all tier initiators; hard blackout; 186/900/500 s Mars-delay/blackout/timeout; 50 seeds.	Same as above.
Weakest-link migration and coordinated relaxation.	Sparse 5/1/1/3; Earth initiator; repeater; hard blackout; 186/900/500 s Mars-delay/blackout/timeout; 50 seeds.	`results/step10/step10_sweep_ci.csv`; `experiments/step10_sweep.py`.
TLA+ verification of quorum intersection.	Exhaustive enumeration, all four tiers $\times$ {strict, k=4, k=3} (27,921 states, complete).	`tla/QuorumIntersection.tla`; `tla/ExhaustiveIntersection.tla`.
Paxos agreement model-checked.	67M states (reduced topology, all-tier Phase 1 family, complete).	`tla/PaxosSmall.tla`.

Table 8: *Claim-to-artifact traceability (continued).*

Claim	Evidence	Artifact(s)
Per-tier capability map: which (Phase 1, Phase 2) capability states each tier holds under each connectivity regime.	Exhaustive per-scenario classification, including the degraded partial-capability states.	`results/capability/capability_map.csv`; `experiments/capability_map.py`.
Containment characterization (Proposition 1) and the $k \leq \lceil \lvert E \rvert / 2 \rceil$ boundary (Proposition 3).	Exhaustive enumeration, 5 $k \times 4$ tiers; containment evaluated as set containment over minimal Q_2, not as the reachability restatement; 0 disagreements. Predicate monotonicity verified, not assumed.	`results/capability/dual_containment.csv`; `experiments/capability_dual_sweep.py`.
Exact four-way gap correspondence and construction-independent audit.	Proposition 1 and Corollary 1 are primary. Independent state enumeration rejects a mutated report; all 129,032 three-node family-pair/pinned-domain cases pass; 16 registered wall cases reconcile with 0 CSV disagreements.	`quorum_audit.py`; `experiments/quorum_audit.py`; `results/capability/quorum_audit_registered.json`.
Wall-specific capability readout.	Deterministic operational demonstration over one planetary and one structurally analogous edge input; not evaluation. Runtime authority remains unknown and service policy is not inferred.	`experiments/capability_readout.py`; `examples/capability/planetary_moon_01.json`; `examples/capability/edge_remote_01.json`.
Uniform threshold corollary: $(1,0)$ reachable iff $q_1 < q_2$; $(0,1)$ iff $q_2 < q_1$.	80 threshold configurations ($n = 3..7$, all valid q_1/q_2), 0 violations.	`results/capability/dual_uniform.csv`; `experiments/capability_dual_sweep.py`.
Behavior inside the two capability gaps: $(1,0)$ matched a healthy contender on recorded metrics and supplied the decided value; $(0,1)$ prevented decision in 50 of 50 seeds when both proposers had an eight-round budget.	Pre-registered single-decree experiment, 5 arms $\times$ 4 incumbent budgets $\times$ 50 seeds; the healthy proposer always had 8 rounds. Predictions, falsification criteria, and the declared consequence of a null were committed before the experiment code existed; two registered predictions were falsified and kept. Output was byte-identical across two runs.	`results/flip/flip_map.csv`; `results/flip/flip_sweep.csv`; `flip.py`; `experiments/flip_sweep.py`; `experiments/flip_verdict.py`.
Price of coincidence: 0 at odd n, 1 at even n; cost-minimal configurations cannot be phase-symmetric at even n.	Re-analysis of the threshold enumeration; post-hoc, not pre-registered, and labelled as such.	`results/capability/dual_uniform.csv`; `experiments/capability_dual_sweep.py`.
Tier-gradient: at $k = 3$, $(1,0)$ is absent at every tier; both gaps are absent for Earth under both readings and for self-reachable LEO. Moon and Mars retain $(0,1)$ at every k; no k closes both gaps everywhere.	Exhaustive enumeration over all 2^{10} connectivity states, every tier and k, under both colocated-acceptor readings. Output byte-identical across two runs.	`results/capability/dual_gradient_map.csv`; `experiments/capability_dual_sweep.py`.
Performance-preserving closure beyond the named wall cases remains open.	Scoped authority, gap-aware behavior, and multi-anchor families are unevaluated candidates, not results.	No artifact; future work.